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SET: Superpixel Embedded Transformer for skin lesion segmentation

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ABSTRACT

Accurate skin lesion segmentation is crucial for the early detection and treatment of skin cancer. Despite significant advances in deep learning, current segmentation methods often struggle to fully capture global contextual information and maintain the structural integrity of skin lesions. To address these challenges, this paper introduces Superpixel Embedded Transformer (SET), which integrates superpixels into the Transformer framework for skin lesion segmentation. Instead of embedding non-overlapping patches as tokens, SET employs an Association Embedded Merging & Dispatching (AEM&D) module to treat superpixels as the fundamental units during both the down-sampling and up-sampling phases. To better capture the multi-scale information of lesions, we propose a superpixel bank to store various superpixel maps with distinct compactness values. An Ensemble Fusion and Refinery (EFR) module is then designed to fuse and refine the results obtained from each map in the superpixel bank. This approach enables the model to selectively focus on different features by adopting various superpixel maps, thereby enhancing the segmentation performance. Extensive experiments are conducted on multiple skin lesion segmentation datasets, including ISIC 2016, ISIC 2017, and ISIC 2018. Comparative analyses with state-of-the-art methods showcase SET's superior performance, and ablation studies confirm the effectiveness of our proposed modules incorporating superpixels into Vision Transformer. The source code of our SET will be available at https://github.com/Wzhjerry/SET.

1. Introduction

Globally, skin cancer ranks among the most prevalent types of cancer. This widespread disease presents a significant health challenge, particularly due to the various forms it can take, such as melanoma, which is known for its high rates of metastasis and severe health risks (Komatsubara et al., 2017; Siegel et al., 2021). Skin cancer manifests itself through various lesions, such as nevus, basal cell carcinoma, and vascular lesions, serving as key biomarkers for different types of skin cancer (Diepgen and Mahler, 2002). Accurate segmentation of these varied skin lesions is crucial, especially melanoma due to its aggressive nature and high prevalence. In dermatology, the early detection and diagnosis of skin cancer are highly dependent on the precise localization of skin lesions. However, manual segmentation is time-consuming and demands a high level of expertise (Zafar et al., 2020). Therefore, automatic skin lesion segmentation techniques are highly needed.

Automatic skin lesion segmentation methods have significantly progressed over time. Initially, hand-crafted features are employed, but they often lack stability and robustness, leading to suboptimal performance. Recent developments in deep learning, especially those in convolutional neural networks (CNNs), have significantly changed the landscape of this field (Ronneberger et al., 2015; Long et al., 2015; Barzegar and Khan, 2022; Gu et al., 2019). Specific models tailored for skin lesion segmentation, such as FAC-Net (Dong et al., 2021), multistage U-Net (Tang et al., 2019), and Ms-Red (Dai et al., 2022), have demonstrated remarkable results. Despite their success, these CNN methods are often limited by their relatively narrow perspective fields. Inspired by the great success of Transformer in both computer vision and natural language processing (Vaswani et al., 2017; Dosovitskiy et al., 2020; Liu et al., 2021), mainly due to Transformer's better handling long-range dependencies with multi-head self-attention modules, researchers have tailored them for medical image segmentation (Chen et al., 2021, 2023). Transformer-based methods specifically designed for skin lesion segmentation, such as Attention Swin-Unet (Aghdam et al., 2023), HiFormer (Heidari et al., 2023), and FAT-Net (Wu et al., 2022), have achieved notable performance. However, Transformer approaches typically process images by dividing

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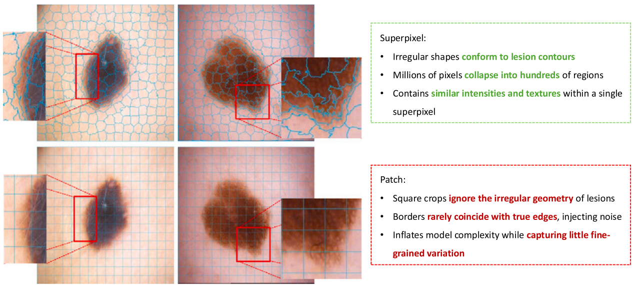

Fig. 1. Representative skin-lesion images partitioned by superpixels and size-fixed square patches. Zoomed-in panels highlight the differences between the two layouts.

them into uniform patches. Although this strategy shows efficacy for natural images, it may not be suitable for dermoscopic images, mainly due to the inherent variable nature of the lesion areas. Simple square-based patch division may result in the loss of crucial prior information on skin lesions, including their semantic and structural attributes. Moreover, these methods still struggle to retain important local features and ensure robustness due to the inflexibility of their patch partitioning strategy. Recognizing this issue, several researchers have proposed to incorporate boundary information into Transformer-based networks to address the limitations (Wang et al., 2021, 2023a; Lin et al., 2023a). Still, these advanced methods treat patches as the fundamental unit.

Superpixels group pixels with similar characteristics, particularly in terms of image intensity and spatial proximity. Functioning as irregular but more homogeneous patches, superpixels carry more information than isolated pixels or regular patches, primarily because they align with object boundaries more closely and parcel images more effectively. This pixel-to-superpixel transformation significantly simplifies an image's complexity, condensing millions of pixels into hundreds or thousands of superpixels with shared attributes. Fig. 1 illustrates representative examples contrasting patch-based partitions versus superpixel partitions. The integration of superpixels into deep learning-based segmentation methods has already validated their utility and effectiveness (Wang et al., 2022; Li et al., 2021; Wang et al., 2023b). Yet, in most of these works, superpixels appear only as an external tool for generating proposals, refining masks, or engineering features, while the backbone still processes fixed square patches or individual pixels. Consequently, the model of interest struggles to learn directly on the boundary-aligned regions, leaving much of the superpixels' descriptive power untapped. A similar pattern holds in volumetric imaging, where supervoxels have proven valuable but are paid less attention (Xie et al., 2024). Viewing superpixels as the fundamental unit for image processing offers several advantages. First, the alignment of superpixels with image boundaries provides a more meaningful and compact representation of features, allowing for more effective capture of local patterns and textures. Second, the generation of superpixels through clustering inherently decreases the number of elements required for processing. Consequently, using superpixels could bring significant advancements in computational efficiency and feature precision, especially when incorporated into Transformer-based models. However, there are two primary challenges: (1) how to effectively integrate superpixels into the Transformer architecture in lieu of patches, and (2) how to effectively exploit the feature representation power of superpixels. A novel framework that naturally embeds superpixels is needed to address these challenges.

In this paper, we introduce SET: Superpixel Embedded Transformer, a novel skin lesion segmentation framework that integrates superpixels into Transformer to improve the skin lesion segmentation performance. By grouping pixels with similar characteristics, superpixels can effectively preserve lesion boundaries and structural details within an image, which are crucial for accurate skin lesion segmentation. Using a superpixel map, SET can map images from pixels to superpixels for being processed within the Transformer framework. We design an Association Embedded Merging & Dispatching (AEM&D) module to process superpixels directly, enhancing computational efficiency and maintaining spatial integrity. Additionally, SET features a'superpixel bank', which archives multiple superpixel-based segmentation results for each to-be-segmented image. Another key component, an Ensemble Fusion and Refinery (EFR) module, consolidates the various semantic information from various superpixel maps into a final output. The core contributions of SET are:

- We propose Superpixel Embedded Transformer (SET) that integrates superpixels into Transformer using two specifically designed modules: AEM&D and EFR. This approach marks the first instance of utilizing superpixels as the fundamental processing unit in Transformer for medical image segmentation.
- We validate the performance of SET on multiple publicly available skin lesion segmentation datasets, including ISIC 2016, ISIC 2017 and ISIC 2018. The evaluation experiments show SET's superior performance, consistently outperforming existing state-of-the-art CNN-based and Transformer-based methods.
- Ablation studies demonstrate the effectiveness and potential of SET, indicating that incorporating more diverse data and exploring various types of superpixel embedding shall further enhance model performance.

2. Related work

2.1. Traditional and CNN-based skin lesion segmentation

In traditional skin lesion segmentation methods, the focus has often been on low-level hand-crafted image characteristics, such as shape, color, and intensity (Silveira et al., 2009). Riaz et al. (2018) introduce an active contour method using the Kullback–Leibler divergence to distinguish between skin lesions and normal skin, aiming to accurately fit a curve at the lesion boundary. Pereira et al. (2019) improve histogram-based methods by incorporating a gradient-based metric to refine histograms, thus achieving better performance over previous contour methods and clustering techniques. Ashour et al. (2018) implement a method using a genetic algorithm to optimize neutrosophic set operations and apply K-means clustering for segmentation. However, hand-crafted features often lack robustness and consistency.

With the advent of deep learning, CNN-based methods have emerged for skin lesion segmentation, outperforming traditional methods that rely on hand-crafted features. Yuan et al. (2017) develop a deep CNN complemented by specific training strategies, such as batch normalization and dropout, and introduce an effective loss function to improve the segmentation performance. Jha et al. (2020) propose DoubleU-Net, which stacks two U-Net architectures and incorporates atrous spatial pyramid pooling to effectively capture semantic and contextual information. FAC-Net (Dong et al., 2021) is equipped with a feedback fusion block and an attention block, enhancing feature extraction and lesion segmentation. Dayananda et al. (2023) propose AMCC-Net that utilizes asymmetric multi-cross convolution and a boundary-guided module for enhanced skin lesion segmentation. Dai et al. (2022) develop Ms-Red, a multiscale residual encoding and decoding network to address challenging cases and improve robustness. Although these CNN-based methods demonstrate significant advancements in skin lesion segmentation, they tend to ignore the crucial aspect of extracting global context features. This neglect limits the ability of these models to fully understand the spatial and contextual relationship within an image of interest, which is essential for accurate skin lesion segmentation.

2.2. Transformer-based skin lesion segmentation

In recent years, Transformer, known for its parallel self-attention mechanism that can effectively capture long-range dependencies, has shown significant advantages in fields such as natural language processing (NLP) and computer vision (Vaswani et al., 2017; Dosovitskiy et al., 2020). Its success has also been notably extended to medical image analysis. For instance, Chen et al. (2021) propose TransUnet, which combines the strengths of both U-Net and Transformer. It uses Transformer as a robust encoder and integrates U-Net to recover localized spatial information, thereby enhancing finer details. Hatamizadeh et al. (2021) introduce SwinUNETR, which employs a Swin Transformer as an encoder. It extracts features at five different resolutions and connects to an FCNN-based decoder at each level via skip connection, ranking the top-performing segmentation approaches.

Transformer has also shown effectiveness in the field of skin lesion segmentation. TransDeepLab (Azad et al., 2022) exploits Hierarchical Swin Transformer with shifted windows to extend DeepLab-v3, incorporating atrous spatial pyramid pooling and showing impressive performance in skin lesion segmentation. DS-TransUnet (Lin et al., 2022) integrates the advantages of Hierarchical Swin Transformer into both its encoder and its decoder to boost the segmentation performance. Beyond purely Transformer-based methods, hybrid ones combining Transformers with CNNs have also shown great performance. ICL-Net (Cao et al., 2022) is designed to understand and model inter-pixel correlations from both global and local perspectives, with an aim of enlarging both inter-class variances and intra-class similarities. TransAttUnet (Chen et al., 2023) features multi-level guided attention and multi-scale skip connection, jointly enhancing semantic segmentation. HiFormer (Heidari et al., 2023) effectively bridges CNN and Transformer extracted features using a Double-Level Fusion module in the encoder–decoder skip connections.

2.3. Prior-guided transformer

Although Transformer has achieved remarkable success in medical image analysis, there exist specific challenges. In particular, its style of segmenting an image into patches for sequence processing may disrupt the inherent semantic and structural integrity of the image. To alleviate this issue, researchers have incorporated boundary-aware prior knowledge into network design. Initial efforts in this direction involve modifying the loss function to emphasize the boundary information, as demonstrated in Boundary Loss (Kervadec et al., 2019). Later, multitask learning approaches are developed, introducing additional branches to provide specific boundary-aware supervision (Lin et al., 2021).

With the evolution of Transformer, the concept of prior knowledge has also been integrated into Transformer's design. A prominent example is the Boundary-aware Transformer proposed by Wang et al. (2021), aiming to address the challenges of significant variations and ambiguities in melanoma's boundaries. Similarly, Lin et al. (2023a) introduce BATFormer, which establishes cross-scale global interactions with reduced computational complexity. It generates flexible windows under the guidance of entropy, effectively addressing the limitations of traditional Transformer models. Additionally, Wang et al. (2023a) propose XBound-Former, a purely attention-based network focusing on boundary knowledge through three specially designed learners. These successful approaches share a common theme: they all emphasize the importance of boundary information, a crucial prior that significantly impacts the skin lesion segmentation performance. Using a Transformer to capture the boundary information, these methods have notably improved the segmentation outputs.

Despite the advancements in Transformer-based segmentation models, it is important to note that these approaches still rely on patches as the fundamental processing unit. However, patches may not effectively capture the contextual and structural information of images, especially lesion boundaries. To address this limitation, researchers have explored alternative processing units that better preserve an image's context and structure. Superpixels have emerged as a promising alternative to patches due to their ability to group pixels into semantically meaningful regions. Several studies have successfully integrated superpixels into Transformer architectures. Zhu et al. (2023) propose the Superpixel Transformer, which leverages superpixels by decomposing the pixel space into a spatially low-dimensional superpixel space. Their model directly predicts a class label for each superpixel and then projects these predictions back into the pixel space for segmentation. Similarly, Mei et al. (2024) introduce SPFormer, which introduces transitions to superpixel representations through a Superpixel Cross Attention module. This module combines local detail preservation with global self-attention, enabling end-to-end trainability while effectively capturing both local and global contextual information. But none of these successful implementations have been applied to the medical imaging domain. Inspired by these works, we here incorporate superpixels within a Transformer architecture and applied it to skin lesion segmentation.

3. Method

In this section, we introduce the overall structure of our proposed SET framework. We then detail two novel modules that leverage superpixels as the fundamental units within Transformer, replacing the traditional patch-based strategy.

Fig. 2 demonstrates the overall framework of SET, which is based on SwinUNETR (Hatamizadeh et al., 2021), a Transformer with an encoder–decoder structure known for its effectiveness in medical image segmentation. The AEM&D module is incorporated into both the encoder and the decoder, which merges and refines features during encoding and decoding. Initially, we generate superpixel maps for all images using the Simple Linear Iterative Clustering (SLIC) algorithm (Achanta et al., 2012). These maps are then transformed into association matrices to facilitate the mapping of pixels to superpixels. Both the original images and their association matrices serve as the network inputs to produce segmentation predictions. We also generate multiple superpixel maps by adjusting the compactness value in SLIC, and store them in a'superpixel bank' for further use. During training, we randomly select superpixel maps from the bank to improve the model's flexibility and robustness. In the inference phase, we employ all available superpixel maps and ensemble the results through the EFR module.

3.1. Merging and dispatching from pixels to superpixels

In Transformer-based segmentation methods, researchers typically employ an encoder–decoder network architecture, with the Transformer serving as the encoder. In conventional Vision Transformers, images are divided into uniformly sized and shaped patches for processing. Unlike conventional patch-based approaches, one of SET's core contributions lies in its utilization of superpixels as the fundamental unit. Applying superpixels brings several advantages: (1) The shape of superpixels is adjustable via specific parameters in SLIC, in contrast to the fixed square shape of patches; (2) They align with the distinct boundaries of the objects of interest, preserving semantic and structural information that regular patches might disrupt; (3) Each superpixel is relatively homogeneous in terms of image intensity, simplifying the processing of individual units, whereas the intensity distribution in a regular patch may be variable.

However, directly applying superpixels as the fundamental unit for processing in Transformers poses challenges.

- The use of relatively fixed convolution layers for patch embedding, effective in standard Transformers, is less practical for SET due to the variable sizes and shapes of superpixels within an image. Additionally, the superpixel maps generated for different images are unique, precluding the use of a standard feature extraction component.

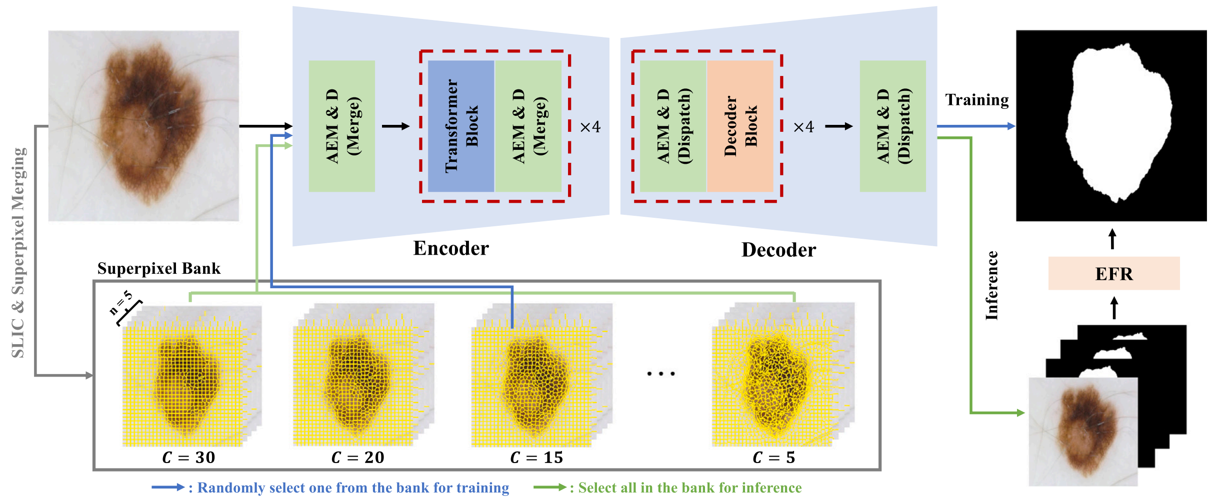

Fig. 2. The overall framework of the proposed SET. It begins by generating superpixel maps for each image using a range of compactness values. These maps are then refined and merged through K-Means clustering at each stage. All generated superpixel maps are stored in the superpixel bank for later use. These superpixel maps will be used for superpixel merging and dispatching through AEM&D. In the training phase, for each training image, a superpixel map is randomly selected from the bank for each epoch. During inference, all superpixel maps are sequentially processed through the model to generate multiple segmentation predictions. The EFR module integrates these predictions along with the original image to produce the final segmentation output.

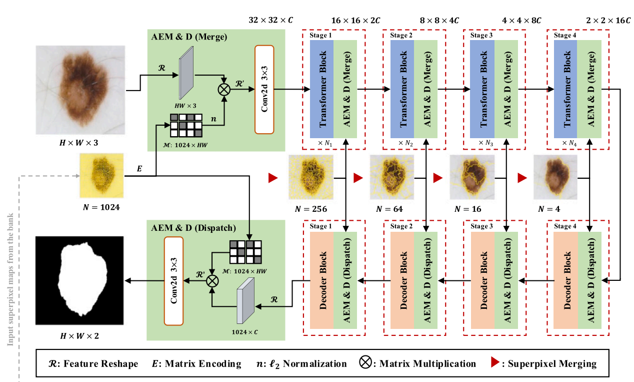

Fig. 3. Detailed structure of the AEM&D module, exemplified with a superpixel count of 1024 per image. We highlight the shape of the output feature at each stage, denoted at the top right corner within each block. At every stage, AEM&D processes a superpixel map, encoding it into an association matrix for subsequent matrix multiplication.

- The standard pooling and upsampling processes shall struggle to adapt to the distinct characteristics and variability of superpixels. Unlike patches, superpixels vary in size and shape within an image. This variability leads to a lack of uniformity and consistency during the pooling and upsampling processes, rendering traditional pooling and upsampling techniques be less effective and less straightforward.

Recognizing these challenges, researchers have typically approached superpixel processing by tokenizing superpixels into features, performing superpixel classification, and then projecting the results back onto pixels using an association matrix (Zhu et al., 2023). In contrast to existing methods, we propose a module named Association Embedded Merging and Dispatching (AEM&D) for simple and effective processing of superpixels within the Transformer framework.

3.1.1. Superpixel embedding

As illustrated in Fig. 3, AEM&D first maps the correspondence between pixels and superpixels, and merges the superpixels using K-Means based clustering. Details of the merging process are presented in Fig. 4. When generating superpixels from an image of size $H \times W$ using the SLIC algorithm, a 2-dimensional map $\mathbf{M}$ of the same size is produced, with values ranging from $[1, n]$, where n represents the number of superpixels generated in the image of interest. The map is then encoded into a binary matrix $\mathcal{M}$ of size $HW \times n$. The process can be represented as

$$\mathcal{M} = \mathbf{O}_n \left[\mathrm{Vec}(\mathbf{M}) - 1 \right], \tag{1}$$

where $\mathcal{M}$ is the resulting embedded matrix of size $HW \times n$, $\mathrm{Vec}(\mathbf{M})$ is a vectorization operation that reshapes matrix $\mathbf{M}$ into a column vector of size $HW \times 1$, $\mathbf{O}_n$ is an embedding operation matrix with size $n \times 1$, which adjusts the values in the vectorized $[\mathrm{Vec}(\mathbf{M}) - 1]$ to be one-hot encoded. The resulting association matrix $\mathcal{M}$ is a binary matrix in which each row corresponds to a superpixel and each column represents a pixel, with a single non-zero value in each column. The nonzero value indicates which superpixel a given pixel belongs to. Each row represents the assignment of pixels for the specific superpixel. This allows us to shift from traditional patch embedding to superpixel embedding, preserving spatial and contextual coherence within the image. The embedding process of the input image can be represented as

$$\mathcal{F}_M(\mathcal{I}, \mathcal{M}) = \mathcal{C}_{3\times3} \left[\mathcal{R}' \left(\mathcal{R}(\mathcal{I}) \times \ell_2 |\mathcal{M}| \right) \right], \tag{2}$$

where $\mathcal{F}_M(\mathcal{I}, \mathcal{M})$ denotes the final output of the embedding process, with $\mathcal{I}$ representing the input image and $\mathcal{M}$ the association matrix. $\mathcal{C}_{3\times3}$ represents convolution with kernel size 3 and stride 3, $\mathcal{R}$ represents the feature reshaping process, where a three-dimensional input feature with size $H \times W \times n$ is reshaped to a two-dimensional feature with size $HW \times n$. Following this, a transpose process is performed to reshape the feature into size $n \times HW$. $\mathcal{R}'$ is the inverse reshape process. The reshaped feature gets multiplied by $\mathcal{M}$ which has a dimension of $HW \times n$. The resulting output feature is of dimension $C \times n$, and the inverse reshape process $\mathcal{R}'$ converts it back to the shape of $H' \times W' \times C$, where $H' \times W' = n$. $\ell_2| \cdot |$ denotes the ℓ_2 normalization operation. The reason why an ℓ_2 normalization is employed during the embedding process is because the original association matrix is a one-hot matrix with only 0 and 1 values in it. Without normalization, multiplying an association matrix sums all features that belong to the same superpixel. The larger the superpixel, the larger the resulting feature magnitudes, which would increase numerical variance and slow convergence, and in extreme cases may lead to training instability. Normalizing each feature to unit l_2 norm removes the size-related bias and keeps the optimization process stable.

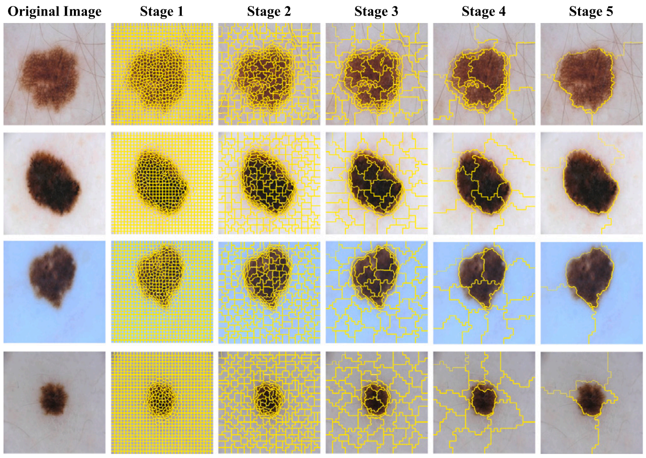

Fig. 4. Representative examples of superpixel merging in five distinct stages, beginning with the initial superpixel embedding and followed by subsequent merging in each Transformer block. The superpixels generated are based on a compactness value of 15.

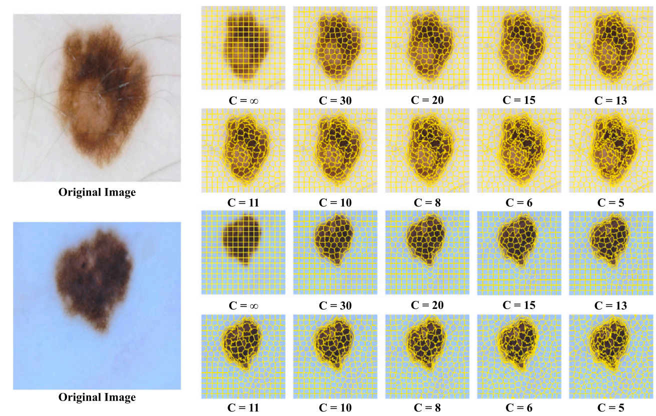

Fig. 5. Representative examples of the generated superpixel maps using different compactness values. Note that C is short for compactness. For better visualization, we set the number of superpixels generated to be 256. The value ∞ represents the original patch separation, serving as an extreme case of superpixel formation.

3.1.2. Merging and dispatching superpixels

The merging and dispatching of superpixels mimic the downsampling and upsampling processes in traditional Transformer-based segmentation methods. For superpixels' merging, we employ K-means clustering to merge the superpixels in a superpixel map by taking the average RGB values and the average positions of pixels within each superpixel as the clustering features. For RGB values, we calculate the average value of each channel of pixels within a superpixel. For the position values, we define the top-left pixel as $(0,0)$ and the pixel at the bottom right corner to be (H,W). Similar to RGB values, we calculate the average position values of all pixels within a superpixel. Together, the clustering feature of K-means is a five-dimensional feature $[\bar{R},\bar{G},\bar{B},\bar{x},\bar{y}]$, where $\bar{R},\bar{G},\bar{B}$ are the mean RGB values and $\bar{x},\bar{y}$ represent the mean coordinates of pixels in the superpixel. The number of clustering centers for the current stage is set to one fourth of that for the previous stage. The merged superpixel map is then used to encode the new association matrix for AEM&D, effectively replacing the standard patch merging with superpixel merging. For instance, the output of AEM&D in the merging process can be represented as the same as that in Eq. (2), except that it takes features from the previous stage f_e and the new encoded association matrix as the input. Similarly, the inverse of the merging process in AEM&D represents the dispatching process in the decoder. It systematically redistributes features from the condensed, merged superpixel representation back to the original pixel level. This transition occurs sequentially: initially from merged superpixels to individual superpixels and subsequently from individual superpixels to pixels. This step-by-step approach ensures that detailed information, potentially lost during the merging phase, can be effectively restored. It also ensures that the dispatched feature retains the original structural and semantic details, which is vital for segmentation. The output of AEM&D in the dispatching process is depicted as

$$\mathcal{F}_D(f_d,\mathcal{M}) = \mathcal{C}_{3\times3}\left[\mathcal{R}'\left(\mathcal{R}(f_d)\times\mathcal{M}^{\mathcal{T}}\right)\right], \tag{3}$$

where $\mathcal{M}$, $\mathcal{R}$, and $\mathcal{R}'$ have the same meanings as those in Eq. (2) and f_d represents the input feature in the dispatching process. The primary distinction in the dispatching process is the use of the transposed association matrix, $\mathcal{M}^{\mathcal{T}}$, to distribute features during decoding.

3.2. Multi-compactness fusion and refinery

Unlike traditional patches that have a fixed size and shape, superpixels can vary in both size and shape within the same image, allowing them to more effectively capture the diverse characteristics of the image of interest. The compactness parameter plays a critical role in this flexibility, dictating the balance between color similarity and spatial proximity during superpixels' generation. Fig. 5 demonstrates how varying compactness values affect the shape and size of the generated superpixel maps, each configuration of which uniquely captures different aspects of the image of interest. Higher compactness values result in more uniformly shaped superpixels by focusing on spatial proximity, improving the model's ability to maintain spatial coherence. Conversely, lower compactness values allow superpixels to align more closely with image contours and boundaries, emphasizing color similarity and capturing fine details. This flexibility enables more detailed and precise extractions of features compared to traditional patch-based approaches.

3.2.1. Application of multiple association matrices

To enhance the model's ability to comprehensively learn semantic information from different superpixel maps, we introduce the concept of a 'Superpixel Bank'. This repository stores multiple superpixel maps generated for a single image, each with a different compactness value. During the training phase, the model randomly selects one of these superpixel maps for image processing, exposing it to a diverse range of representations. In the inference stage, the model processes all available superpixel maps for a given image. By integrating the semantic information from all these maps, the model achieves greater accuracy in estimating lesion areas. This strategy enriches the model's learning experience by enabling it to understand various aspects of the image without the need to select a single optimal superpixel map. As a result, it enhances the model's ability to capture and interpret variations in intensity and structure that characterize the images. The use of multiple superpixel maps also addresses inconsistencies across different images, ultimately improving the overall performance.

3.2.2. Segmentation fusion and refinery

In the inference stage of our model, we utilize all the superpixel maps from the Superpixel Bank to generate multiple segmentation outputs for a single image. Each of these outputs may emphasize different aspects of the skin lesion, reflecting the variability that arises from using diverse superpixel maps. To effectively combine the information captured by these various maps, it is essential to integrate the segmentation results. While existing methods like Gradient Boosting, Random Forest, or Majority Voting can be used for this purpose, they often struggle in complex scenarios, particularly when trying to accurately identify boundary regions with conflicting predictions from different superpixel maps. To address this challenge, we develop the Ensemble Fusion and Refinery (EFR) module. EFR is specifically designed to adeptly handle the intricacies of integrating segmentation results

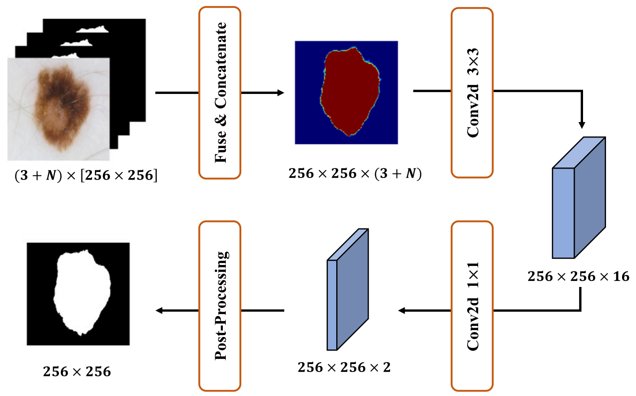

Fig. 6. Detailed architecture of the proposed EFR, with notations indicating the shape of the features at each step.

from multiple superpixel maps, effectively addressing complexities and ambiguities within the image. By leveraging all available superpixel maps, this approach ensures more robust and accurate segmentation, producing a more reliable output.

Fig. 6 illustrates the detailed structure of our proposed EFR module, which operates in three primary steps. First, the EFR module receives all individual segmentation results along with the original image, fusing them to create a composite feature set. This fusion step is critical for integrating the distinct information provided by each segmentation output. Next, the fused features are processed through two convolutional layers, enhancing the accuracy and reliability of the combined features. Finally, the processed features undergo post-processing, which includes techniques like hole filling and largest connected component analysis, refining the output to ensure it is coherent and practically applicable. The training process of EFR involves using the trained segmentation network to generate segmentation outputs for each image in the dataset using the available superpixel maps. These segmentation results, along with the original image, constitute the input data to the EFR module. The training process of EFR follows standard deep learning frameworks, with the distinction lying in the unique nature of the input.

4. Implementation details

4.1. Datasets

The proposed SET is evaluated on three datasets for skin lesion segmentation from the International Skin Imaging Collaboration (ISIC) Challenge, each offering a unique set of dermoscopic images: ISIC 2016 (Gutman et al., 2016), ISIC 2017 (Codella et al., 2018) and ISIC 2018 (Codella et al., 2019).

- ISIC 2016: The dataset comprises a total of 1279 dermoscopic images, which are officially divided into 900 images for training and 379 images for testing. We extract a subset of 180 images from the training set as the validation set in our experiments.
- ISIC 2017: The dataset features 2750 dermoscopic images, which are officially divided into a training set of 2000 images, a validation set of 150 images, and 600 images for testing. The dataset split in our experiments adopts this official configuration.
- ISIC 2018: The dataset includes 2594 dermoscopic images. In our experimental setup, we adopt a randomized selection approach, allocating 64% of the images for training, 16% for validation, and 20% for testing, following previous works (Dai et al., 2022; Wu et al., 2022; Wang et al., 2023b).

In addition to these official ISIC datasets, we expand our scope of research by aggregating a diverse array of datasets directly from the official ISIC Archive website using the provided API.[2] This collection encompasses a total of 13,760 dermoscopic images, with corresponding segmentation labels. For this largest dataset, we randomly split it into a training set (64%), a validation set (16%) and a testing set (20%).

4.2. Network architecture

Our SET implementation is built upon the SwinUNETR architecture (Hatamizadeh et al., 2021), utilizing the Swin Transformer as the encoder and a custom-designed decoder. We set the patch embedding size to 8 and the window size to 4, corresponding to dividing the input image into 1024 superpixels. The encoder comprises layers with depths [2, 2, 18, 2], and the number of attention heads for each layer is set to [4, 8, 16, 32], respectively. Both the encoder and the decoder weights are randomly initialized.

We integrate the AEM&D module into both the encoder and decoder. In the encoder, the AEM&D module in its merging state replaces the original patch embedding and patch merging components, including the initial patch embedding layer. In the decoder, the AEM&D module in its dispatching state substitutes the original transpose convolution layers used for feature upsampling. The EFR module is implemented before the final output of the network. Except for the fuse & concatenate module, and post-processing module, we implement a 3×3 convolution with a kernel size of 3 and strides of 3, followed by a 1×1 convolution with a kernel size of 1 and strides of 1.

To implement the association matrix in the AEM&D module, we first allocate a zero-filled tensor with the target matrix dimension and register it as a non-trainable buffer. At every training step, the precomputed association matrix is updated into this buffer, and is then used exclusively in the merging and dispatching operations. Because the matrix itself contains no learnable weights, it is excluded from gradient computation and contributes nothing to back-propagation nor parameter updating. This design helps minimize memory overhead while ensuring that the association information is consistently available throughout training.

Post-processing consists of two standard, non-parametric morphological steps applied uniformly to every model, including the baseline: (1) hole filling, implemented via morphological closing followed by flood-fill to remove empty regions inside lesions, and (2) largest connected component filtering, which suppresses isolated false positives and preserves only the dominant lesion. These are conventional morphological post-processing steps that operate solely on the spatial connectivity of foreground pixels and do not involve any tunable parameters.

4.3. Implementation

The images are resized to 256×256. Online data augmentations, including color jittering, Gaussian noise addition, random affine transformation, and Gamma enhancement, are employed to mitigate overfitting. For superpixel map generation, the SLIC algorithm is utilized with varying compactness values ranging from 5 to 30.

SET's implementation utilizes PyTorch (Paszke et al., 2019) operated on a workstation equipped with NVIDIA RTX A6000 GPUs. In the training phase, the AdamW optimizer is used with an initial learning rate of 0.05 and the weight decay to be 0.01. The learning rate progressively decays following a cosine scheduler during training. The batch size is 48. The training objective is a weighted combination of Cross Entropy Loss and Dice Loss, with weights of 1.0 and 1.5, respectively. Training is conducted over 100 epochs, and we select the model that achieves the best performance on the validation set for testing. For evaluation, we employ the Dice Similarity Coefficient (DSC), Jaccard Index (JAC), and Accuracy (ACC) as metrics to assess the segmentation performance.

[2] https://api.isic-archive.com/collections/.

Table 1
Quantitative comparisons of SET with other SOTA skin lesion segmentation methods on ISIC 2016. Note that $\mathcal{C}$ is for CNN-Based methods and $\mathcal{T}$ is for Transformer-based methods. The best results are highlighted in **bold**.

Type	Method	JAC ↑	DSC ↑	ACC ↑
$\mathcal{C}$	16'Team EXB	0.8430	0.9100	0.9530
	19'CE-Net (Gu et al., 2019)	0.8591	0.9190	0.9596
	21'FAC-Net (Dong et al., 2021)	0.8623	0.9251	0.9609
	22'Ms-Red[a] (Dai et al., 2022)	0.8344	0.8996	0.9457
$\mathcal{T}$	21'TransUnet[a] (Chen et al., 2021)	0.8583	0.9135	0.9557
	22'FAT-Net[a] (Wu et al., 2022)	0.8522	0.9125	0.9578
	23'APFormer[a] (Lin et al., 2023b)	0.8559	0.9141	0.9560
	23'BATFormer[a] (Lin et al., 2023a)	0.8566	0.9157	0.9583
	SET (Ours)	**0.8697**	**0.9270**	**0.9638**

[a] Self-implementation.

Table 2
Quantitative comparisons of SET with other SOTA skin lesion segmentation methods on ISIC 2017. Note that $\mathcal{C}$ is for CNN-Based methods and $\mathcal{T}$ is for Transformer-based methods. The best results are highlighted in **bold**.

Type	Method	JAC ↑	DSC ↑	ACC ↑
$\mathcal{C}$	17'Team Yuan	0.7650	0.8490	0.9340
	19'CE-Net (Gu et al., 2019)	0.7754	0.8561	0.9315
	21'FAC-Net (Dong et al., 2021)	0.7427	0.8491	0.9363
	22'Ms-Red[a] (Dai et al., 2022)	0.7632	0.8483	0.9310
$\mathcal{T}$	21'TransUnet[a] (Chen et al., 2021)	0.7540	0.8436	0.9311
	22'FAT-Net[a] (Wu et al., 2022)	0.7692	0.8501	0.9352
	23'APFormer[a] (Lin et al., 2023b)	0.7680	0.8528	0.9331
	23'BATFormer[a] (Lin et al., 2023a)	0.7699	0.8530	0.9358
	SET (Ours)	**0.7791**	**0.8599**	**0.9388**

[a] Self-implementation.

5. Experiments

5.1. Comparisons with SOTA

In this section, we compare SET with other SOTA skin lesion segmentation methods covering two main categories: CNN-based methods including CE-Net (Gu et al., 2019), FAC-Net (Dong et al., 2021), and Ms-Red (Dai et al., 2022), and Transformer-based methods including TransUnet (Chen et al., 2021), FAT-Net (Wu et al., 2022), APFormer (Lin et al., 2023b), and BATFormer (Lin et al., 2023a). For ISIC 2016 and ISIC 2017, we also include the top performing methods from the respective ISIC challenge leaderboards in 2016[3] and 2017.[4] The ISIC 2018 comparison involves only SET and other SOTA methods, as the test set used for these SOTA methods is separated from the dataset, which is distinct from the official test set used for leaderboard submissions.

5.1.1. Quantitative results

Tables 1, 2, and 3 provide the quantitative comparisons of our SET against other SOTA methods on the ISIC 2016, 2017, and 2018 datasets. We reimplement Ms-Red, TransUnet, FAT-Net, APFormer, and BATFormer based on their official repositories. The results for CE-Net and FAC-Net are directly copied from their original papers. Our SET demonstrates superior performance on all three datasets. Specifically, on ISIC 2016, SET improves the JAC score to 0.8697 compared to the previously best method, BATFormer. On the ISIC 2017 dataset, SET attains the highest scores across all evaluation metrics. Furthermore, on ISIC 2018, SET achieves a 2.03% improvement in JAC compared to BATFormer, underscoring its effectiveness across varied datasets. The consistent superior performance of SET across different datasets highlights its robustness and adaptability to various data characteristics and segmentation challenges.

5.1.2. Qualitative results

Fig. 7 provides qualitative comparisons between SET and other SOTA methods on the ISIC 2016, 2017, and 2018 datasets. The segmentation results are overlaid on the dermoscopic images with cyan boundaries for visual comparisons. These results demonstrate that SET achieves more precise segmentation, particularly in boundary regions and areas that are challenging to differentiate. This improved accuracy suggests that employing superpixels enables the model to learn diverse semantic information by analyzing different regions. The enhanced segmentation outputs also highlight the advantage of using multiple superpixel maps in SET. By ensembling different regions for analysis, the model captures diverse semantic information, contributing to its overall accuracy.

[3] https://challenge.isic-archive.com/leaderboards/2016/.
[4] https://challenge.isic-archive.com/leaderboards/2017/.

Table 3
Quantitative comparisons of SET with other SOTA skin lesion segmentation methods on ISIC 2018. Note that $\mathcal{C}$ is for CNN-Based methods and $\mathcal{T}$ is for Transformer-based methods. The best results are highlighted in **bold**.

Type	Method	JAC ↑	DSC ↑	ACC ↑
$\mathcal{C}$	19'CE-Net (Gu et al., 2019)	0.8282	0.8959	0.9597
	21'FAC-Net (Dong et al., 2021)	0.8399	**0.9119**	0.9641
	22'Ms-Red[a] (Dai et al., 2022)	0.8345	0.8999	0.9619
$\mathcal{T}$	21'TransUnet[a] (Chen et al., 2021)	0.8124	0.8889	0.9529
	22'FAT-Net[a] (Wu et al., 2022)	0.8202	0.8903	0.9578
	23'APFormer[a] (Lin et al., 2023b)	0.8256	0.8968	0.9582
	23'BATFormer[a] (Lin et al., 2023a)	0.8261	0.8960	0.9601
	SET (Ours)	**0.8464**	0.9098	**0.9645**

[a] Self-implementation.

Table 4
Quantitative analysis of component contributions in SET. For entries in the second and third rows, where multiple superpixel maps are used as inputs without EFR, the final metrics represent the average values derived from metrics across different superpixel maps.

Method	JAC ↑	DSC ↑	ACC ↑
Baseline	0.8046	0.8754	0.9500
Baseline + AEM	0.8093	0.8828	0.9560
Baseline + AEM&D	0.8290	0.8952	0.9609
Baseline + AEM&D + EFR (SET)	0.8464	0.9098	0.9645

5.2. Effectiveness of components in SET

To quantify the contributions of the AEM&D and EFR modules in our SET framework, we perform ablation experiments on the ISIC 2018 dataset. Throughout these tests, the superpixel count is fixed at 1024 with 10 superpixel maps. Since the EFR module operates on multiple outputs that are only available when there are multiple superpixel maps, it cannot be separately evaluated. The results in Table 4 reveal three key findings. First, adding the AEM module—even without its dispatching branch—already improves segmentation accuracy, suggesting that AEM alone enhances feature representation. Second, enabling the position-aware dispatching component further improves semantic understanding and structural capturing, demonstrating the complementary effect of the full AEM&D configuration. Third, incorporating the EFR module significantly boosts performance by integrating complementary information from multiple superpixel maps, highlighting EFR's role in refining boundaries where diverse spatial perspectives are integrated. Collectively, these findings confirm that each module offers distinct yet complementary benefits, and that their combined use yields the best segmentation performance.

5.3. Normalization scheme comparison

In this section, we investigate how different normalization strategies for the association matrix affect the segmentation performance. In addition to our default ℓ_2 normalization, we also evaluate ℓ_1 normalization.

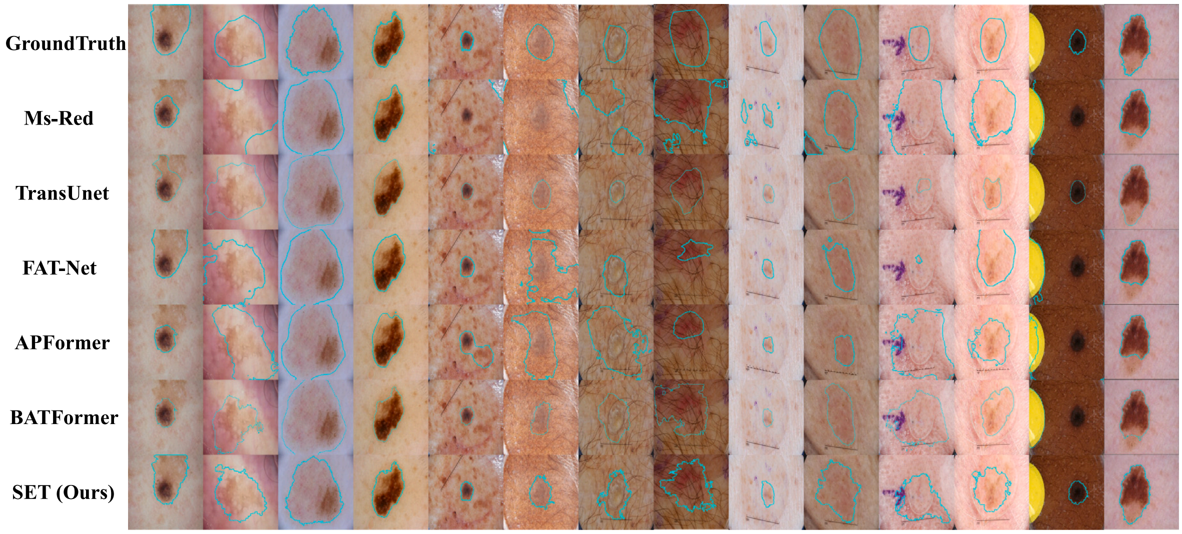

Fig. 7. Qualitative segmentation comparisons of SET, BATFormer, APFormer, FAT-Net, TransUnet and Ms-Red. The segmentation results are overlaid on the dermoscopic images with cyan boundaries.

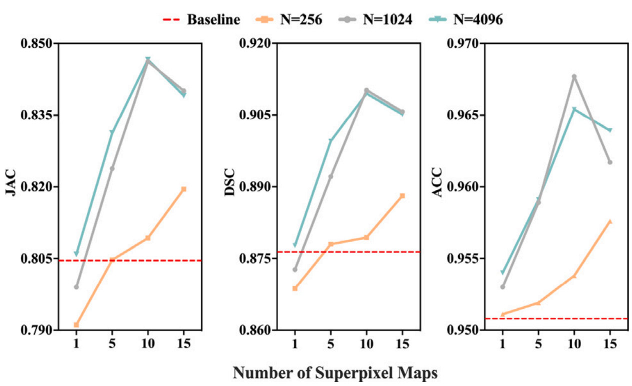

Fig. 8. Skin lesion segmentation performance of SET using different numbers of superpixel counts and superpixel maps. The *x*-axis shows the number of superpixel maps with different compactness values. The *y*-axis denotes different evaluation metrics. The red dotted line represents the baseline performance.

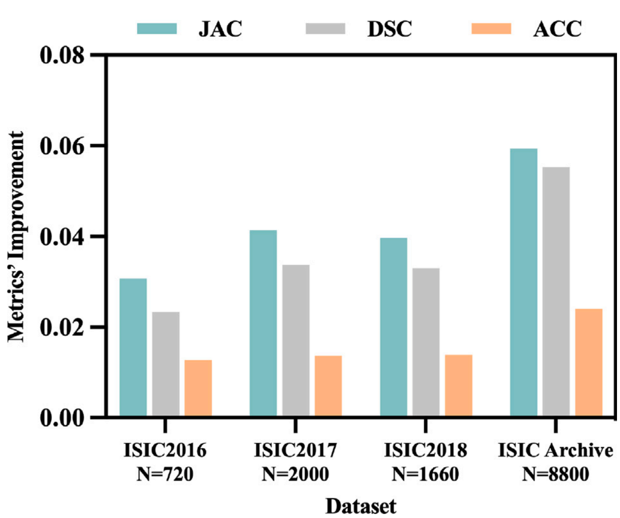

Fig. 9. Improvements of SET over the baseline in terms of three evaluation metrics using datasets with different numbers of training images.

Table 5
Performance of SET under different normalization schemes.

Method	JAC ↑	DSC ↑	ACC ↑
Baseline	0.8046	0.8754	0.9500
ℓ_1 normalization	0.8212	0.8902	0.9585
ℓ_2 normalization	0.8464	0.9098	0.9645

The comparison results are tabulated in Table 5. As shown in that table, ℓ_1 normalization yields inferior performance across all evaluation metrics compared with ℓ_2 normalization. This may be attributed to the fact that ℓ_1 normalization tends to induce sparsity by driving some feature components toward small values, thereby reducing the expressive capacity of the token embeddings. In contrast, ℓ_2 normalization preserves the relative structure of all components while bounding their overall magnitudes, resulting in more stable gradients, faster convergence, and improved generalization. For these reasons, we adopt ℓ_2 normalization as our default setting in all experiments.

5.4. Training efficiency and memory footprint

This section analyzes the training efficiency and memory cost of SET. Introducing superpixel bank adds certain computational overhead, but its impact on the overall efficiency is modest for three reasons. (1) Pre-computation: All superpixel maps, including those generated with varying compactness values as well as their hierarchical K-means clustering results, are computed only once and are derived offline and cached. As a result, they impose no memory pressure nor runtime overhead during training. (2) Training-time usage: For each image during training, we randomly select a single superpixel map from the bank. This map is used only in the merging and dispatching processes, which are implemented as batched matrix multiplications and thus execute efficiently in parallel on GPU, introducing negligible additional computation. (3) Validation: During evaluation, each image is processed multiple times for different superpixel maps, with all maps updated to GPU memory to ensure that switching between them introduces virtually limited overhead. This design has a similar effect as increasing the effective batch size without significantly prolonging the inference time. Alternative traditional methods, such as majority voting based ensembling, fail to match the accuracy achieved by our superpixel bank approach (0.8111 versus 0.8464 in JAC), while incurring comparable runtime costs. Table 6 reports the total training time, average per-image inference time and model parameters using different superpixel bank

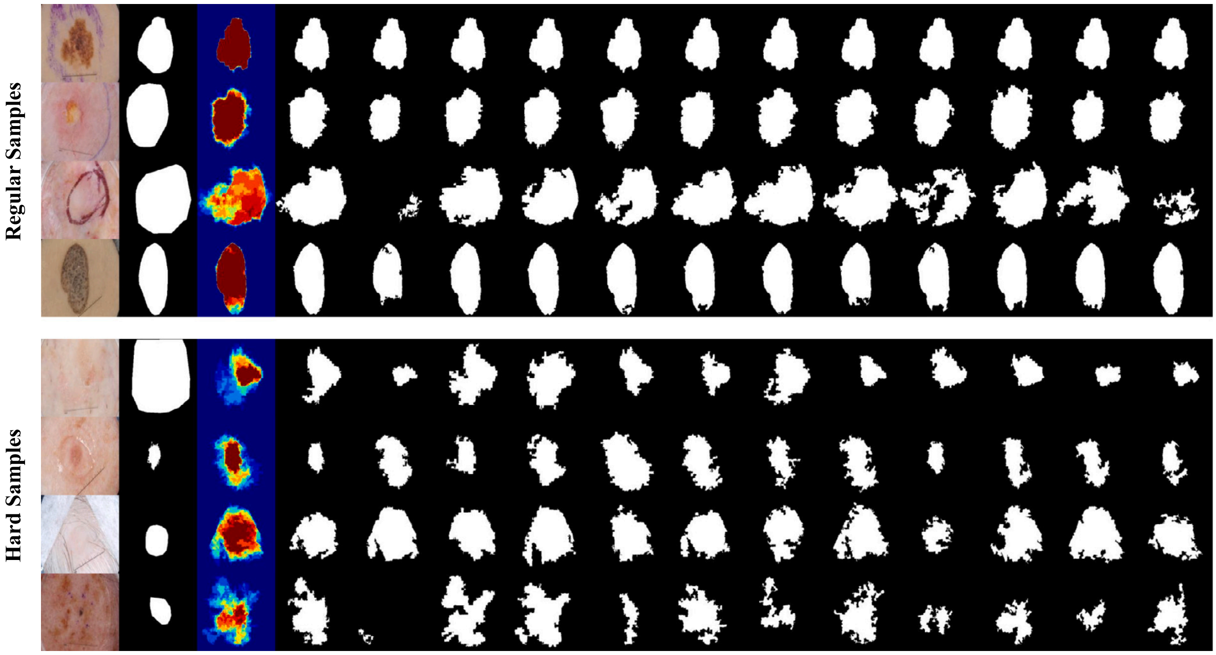

Fig. 10. Representative segmentation results using different superpixel maps and the combined heatmap of all the segmentation predictions. Displayed from left to right are the original image, groundtruth, merged output heatmap, final segmentation prediction and outputs generated using different superpixel maps. The examples are categorized into 'regular' and 'hard'.

Table 6
Comparisons of total training time, average per-image inference time, and model parameters of SET under different superpixel bank sizes. All experiments are conducted on an NVIDIA RTX A6000 GPU. The batch size is set to 4 during inference.

Size of bank	Training time	Inference time	Param.
Baseline	9.2 h	0.08 s	230M
5	13.1 h	0.12 s	275M
10	13.5 h	0.14 s	275M
15	14.2 h	0.18 s	275M

sizes. Due to an optimized deployment implementation of the association matrix, the inference time using different numbers of superpixel maps does not scale linearly with the superpixel bank size. The results indicate that even with a ten-map bank, the model parameters remain unchanged, and the total training time increases by only about 5% compared to the five-map setting, confirming that the accuracy gains introduced by the superpixel bank come at a minimal cost.

5.5. Impact of superpixel count and map variability

In this part, we systematically evaluate how the superpixel count per image and the number of superpixel maps used influence performance. The experiment includes setting the superpixel counts as 256, 1024, and 4096 within a fixed image dimension framework. Additionally, we vary the number of superpixel maps employed, examining configurations encompassing 1, 5, 10, and 15 maps. Different superpixel maps are generated using varying compactness values, systematically sampled within the range from 5 to 30. As shown in Fig. 8, the combination of 1024 superpixels and 10 superpixel maps yields the best performance.

The enhanced performance of SET, as compared to the baseline represented by the red dotted line, demonstrates the superiority of employing superpixels over traditional patches. This result highlights the effectiveness of SET in leveraging the characteristics of superpixels for improved skin lesion segmentation. In particular, when the superpixel count is set to 1024 or 4096, the segmentation performance improves as the number of superpixel maps increases from 1 to 10 but slightly decreases with 15 maps. We conjecture that a larger number of ensemble requires the EFR module to merge a substantially larger set of candidate masks. With each added map, the model must handle an extra token stream and incorporate it into the final representation. Because our fusion head is deliberately lightweight, its capacity scales sub-linearly with the number of inputs. Furthermore, all maps are generated with compactness values uniformly sampled from the empirically validated interval [5, 30]. Ten values sampled from this range yield qualitatively distinct superpixel maps that span the spectrum from highly regular to highly adaptive partitions. As the size of the bank grows, successive samples become increasingly similar, and the structural differences across maps diminish. The primary function of the bank is to expose the network to diverse superpixel maps. Beyond ten maps, the marginal gain in variability is negligible, while the model still needs to allocate capacity to process each new map, increasing the risk of overfitting to a varying set.

Furthermore, during the generation of superpixel maps, we note that with a superpixel count of 256 and smaller compactness values, the resulting superpixels are highly complex and irregular, posing significant challenges for the superpixel embedding process. The empirical results when the superpixel count is set to 256 corroborate this observation. While the performance differs between using 1024 and 4096 superpixels, there is no statistically significant difference in their performance. However, the computational cost associated with a superpixel count of 4096 is considerably higher than that for 1024. Consequently, for subsequent experiments, we standardize the superpixel count as 1024 and the number of superpixel maps as 10 to balance performance and computational efficiency.

5.6. Cross-dataset evaluation compared to baseline

We here examine how the variation in the size of the training dataset affects the performance of the proposed SET model comparing to the baseline. We utilize four datasets with different training set sizes of 720, 2000, 1660, and 8800, respectively. Fig. 9 presents the improvements of SET's performance across all evaluation metrics relative to the baseline.

The results demonstrate that SET consistently outperforms the baseline across different datasets with various sizes of training sets. Generally, SET achieves better performance as the size of the training dataset increases, underscoring the scalability of SET and the potential for further improvement with more data. To further validate these findings, Student's t tests are conducted to compare the performance of SET with that of the baseline across the various dataset sizes. The tests reveal statistically significant differences ($p < 0.001$) in all cases, confirming that the performance gains achieved by SET are substantial. This robust cross-dataset performance highlights SET's effectiveness and reliability compared to the baseline, regardless of the training dataset's size.

6. Discussions

This discussion highlights the effectiveness of using multiple superpixel maps in our SET for the task of skin lesion segmentation. Fig. 10 presents heatmaps that represent the combined segmentation predictions using different superpixel maps. These heatmaps serve as a visual tool to illustrate the impact of generating multiple predictions. The heatmaps clearly show that different superpixel maps force the model to focus on different regions of the image, leading to diverse segmentation predictions. In standard cases, these variations mainly affect the boundary regions, prompting the model to capture different features of the lesions. Although individual predictions from a single superpixel map may not be comprehensive, the combined prediction from all superpixel maps results in more accurate segmentation.

Challenges emerge with more complex cases, as shown in the figure, where the predictions from different superpixel maps show significant variance. These observations underscore the difficulty of achieving a coherent final segmentation. Our EFR module plays a critical role in this context, effectively merging the varied predictions into a unified and more accurate segmentation result. This approach demonstrates that leveraging multiple superpixel maps enables the model to analyze the image from different perspectives, ultimately enhancing the overall segmentation quality. This strategy is particularly effective in complex cases, showing the potential of superpixel-based methods to improve the accuracy of medical image segmentation.

7. Conclusion

In this study, we introduce the Superpixel Embedded Transformer (SET), a novel approach designed to address the limitations of traditional patch-based Transformer methods in segmenting skin lesions. SET uniquely embeds superpixels into the Transformer framework to enhance the analysis of dermoscopic images, capturing both global context and local details. It employs superpixel maps to transform images from pixels to superpixels and utilizes specialized modules, Association Embedded Merging & Dispatching (AEM&D) and Ensemble Fusion and Refinery (EFR), to improve feature extraction and segmentation accuracy. Extensive tests on major skin lesion datasets have confirmed SET's effectiveness, demonstrating its promise for dermoscopic image analysis. Looking ahead, we plan to expand SET in three directions. First, we will devise faster, more memory-efficient pixel-to-superpixel mapping and explore denser association representations to shorten preprocessing time and further improve computational efficiency. Second, we plan to extend SET from 2D dermoscopic images to 3D volumes, enabling direct analyses of CT, MRI, and other volumetric scans where supervoxels can capture spatial continuity across slices. Third, we will adapt the proposed framework to multimodal imaging, for example, fusing dermoscopy with clinical photography or combining MR and PET, to evaluate SET's ability to integrate information from complementary modalities. Collectively, our future work aims to generalize the benefits of superpixel embedding beyond skin-lesion segmentation and make SET a versatile tool for a broad range of medical-imaging applications.

CRediT authorship contribution statement

Zhonghua Wang: Validation, Project administration, Writing – original draft, Resources, Writing – review & editing, Methodology. **Junyan Lyu:** Writing – review & editing, Methodology, Validation. **Xiaoying Tang:** Supervision, Writing – review & editing, Funding acquisition.

Declaration of competing interest

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Data availability

I have shared the link to my data and code in the manuscript.

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